📘 **Revision Booklet: Science Exam 2 (Lessons 7–9)**
Name: _____ **Class:** _____

✅ **Exam Date: Thursday**

📌 **Topics Covered:**

1. **Energy and You** (Lesson 7)
2. **Fuels** (Lesson 8)
3. **Renewable Resources** (Lesson 9)

🔋 **Lesson 7: Energy and You**

Key Concepts to Know:

- What energy is and the **law of conservation of energy**
- Different types of energy:
  - Chemical
  - Kinetic
  - Thermal
  - Gravitational potential
- How the body uses energy from food
- Energy transfer and storage in the body
- Energy loss as heat

Questions to Check Understanding:

1. Define energy.
2. What are the four main types of energy used by the body?
3. Describe the energy transformations that occur when you go for a run.
4. Why is chocolate a quick source of energy?
5. What happens to excess energy in the body?

Revision Task:
📝 Draw a diagram showing how energy from food is transformed and used in the human body.

🔥 **Lesson 8: Fuels**

Key Concepts to Know:

- What a fuel is
- How fossil fuels (coal, oil, natural gas) were formed
- What **combustion** is and what it releases
- Uses of fuels (e.g., transport, electricity, heating)
- Environmental effects of burning fuels (e.g., CO_2 and climate change)
- Definition of **non-renewable**

Questions to Check Understanding:

1. What are fossil fuels and how are they formed?
2. What is combustion and what are its products?
3. Name three uses of fossil fuels.
4. Why are fossil fuels non-renewable?
5. How does burning fossil fuels contribute to climate change?

Revision Task:
📊 Create a table listing **Coal**, **Oil**, and **Natural Gas** — include how each is formed, what it's made from, and a key use.

🌱 Lesson 9: Renewable Resources

Key Concepts to Know:

- What renewable resources are and why they're important
- Examples: Solar, Wind, Hydroelectric, Geothermal, Biomass
- Pros and cons of each energy type
- Why some energy sources are better for certain locations
- The challenge of using renewables everywhere

Questions to Check Understanding:

1. What does “renewable” mean?
2. Name and describe 5 renewable energy sources.
3. Which renewable source works best in rainy areas?
4. What are the main pros and cons of solar and wind power?
5. Why are governments investing more in renewable energy?

Revision Task:
🧠 Draw a Venn diagram comparing **Renewable** and **Non-Renewable** energy sources: overlap should show similarities.

📝 Final Practice

Short Answer Practice Questions:

1. Why is geothermal considered a reliable energy source?
2. Describe how hydroelectric power generates electricity.
3. Which renewable source still releases CO_2, and why?
4. What are two reasons some countries don't use solar power?
5. What is the environmental benefit of switching from fossil fuels to renewables?

Challenge:
Write a paragraph explaining why no single renewable energy source is perfect for all locations.

✅ Revision Checklist:

Revision Task	Done?
Reviewed Canvas Slides for Lessons 7–9	☐
Completed all "Check Understanding" questions	☐
Made diagrams/tables for key ideas	☐
Practiced short-answer questions	☐
Asked questions if unsure	☐